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RESEARCH PROPOSAL

Proof-Gated Medication Safety: A Neuro-Symbolic Pipeline with Machine-Checkable Certificates for Clinical Text

Research Topic: Verifiable, explainable AI for medication-safety checking in clinical text (drug–drug interactions, contraindications and renal-dosing rules)

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Introduction:

Large language models (LLMs) and other neural systems can read clinical text and suggest actions, but they can produce statements that sound confident yet are not supported by evidence [1]. In medicine this is dangerous, because medication errors are a leading cause of avoidable harm in health care [2], and regulation such as the EU AI Act places transparency and human-oversight obligations on high-risk AI systems, including many clinical applications [3]. This project therefore proposes a system with two parts that work together:

1) **The neural part (AI):** It reads the patient's notes, extracts key facts such as age, kidney function and current medicines, and proposes a conclusion, for example "these two medicines should not be taken together."
2) **The symbolic part (rule-based):** It checks the AI's conclusion against a trusted, expert-built database of medical rules. It accepts the conclusion only if it can be proven step by step from those rules, and a separate checking program then verifies the proof independently.

The main focus is medication safety, covering drug–drug interactions, contraindications and renal-dosing rules. The project delivers an open-source framework, an expert-annotated benchmark and an empirical study of where the neural-symbolic interface fails.

The gaps: Existing methods fall short. AI that looks information up does not verify that its answer follows from what it found, and other neuro-symbolic systems are mostly tested on simple, artificial problems. This project aims to fill these gaps with proof-backed answers that are clear about their limits. Rather than treating silence as safety, the system separates "no known risk" (the knowledge base was checked and nothing was found) from "the knowledge base has no information," and its outputs are judged by faithfulness measures and human experts.

Research Questions and Hypotheses:

Q1. Can certificate-gating reduce unsupported or unsafe medication-safety claims made by LLMs, and at what cost in accuracy and coverage?

Q2. Are certificate-based explanations more faithful than post-hoc LLM rationales?

Q3. How do errors in neural fact extraction and knowledge-base gaps propagate through the pipeline, and can the system detect its own fragility?

Q4. Do certificates improve pharmacists' ability to catch errors, or do they induce over-trust?

H1. **Fewer unsupported claims.** At matched coverage of at least 80% of cases, certificate-gated output reduces the rate of unsupported safety claims by at least 50% (relative) compared with the same LLM without verification.

H2. **Small loss in accuracy.** Gating costs no more than 2 F1 points on interaction and contraindication detection relative to the best neural baseline (F1 combines how many dangers are found with how few false alarms are raised), with an abstention rate of 20% or less.

H3. **Grounded and faithful proofs.** At least 95% of the premises in accepted certificates are grounded in a cited evidence span in the source note, and certificate-based explanations outperform LLM-generated rationales on a perturbation-based faithfulness test, in which the input is deliberately changed to see whether the explanation reflects what drove the answer.

H4. **Proofs help pharmacists.** Pharmacists reviewing certificates detect deliberately injected errors at a higher rate than those reviewing free-text rationales, without a significant increase in acceptance of incorrect recommendations (that is, without creating false confidence).

Literature Review:

Explainability is a stated requirement in safety-critical domains, yet common explanation methods have a known weakness. Jacovi and Goldberg separate plausibility (an explanation that sounds

18 reasonable) from faithfulness (an explanation that reflects the real reasoning) [4], and Rudin argues that high-stakes decisions call for interpretable models rather than post-hoc explanations of black boxes [5]. Experiments support this concern: chain-of-thought rationales can leave out the real influences on a model's answer [6], and their faithfulness changes with the model and the task [7].

Retrieval-augmented generation and graph-based variants such as GraphRAG give an LLM external knowledge to read [8], [9]. They supply context, but they do not check that the generated 1 answer actually follows from it. Neuro-symbolic frameworks such as DeepProbLog [10], NeurASP [11], Scallop [12] and Logic Tensor Networks [13] combine neural perception with logical reasoning, but they are mostly evaluated on synthetic or perception benchmarks, rarely on messy clinical text with incomplete knowledge bases. Concept bottleneck models [14] expose intermediate concepts, but they do not derive conclusions from an external, auditable rule base.

Reading medication information from clinical notes is well studied, for example in the n2c2 2018 shared task [15], and errors made at this step are known to pass on to later stages. What is still missing is a clinically grounded pipeline that (a) produces a machine-checkable certificate for each accepted output, (b) is explicit about what the certificate does and does not guarantee, (c) distinguishes "no known risk" from "the knowledge base is silent," and (d) is evaluated with faithfulness metrics and human experts rather than plausibility alone. This project is designed to fill that gap.

Proposed Methodology:

1. **Research Design:** We will use a mixed-methods design with four linked parts: (i) building the proof-gated framework, (ii) building an expert-annotated benchmark, (iii) an automated comparison of gated and ungated systems to test H1–H3, and (iv) a within-subject study with pharmacists to test H4. Failure cases from all parts will be analysed qualitatively to answer Q3.

   Data and knowledge base: The benchmark will contain about 600 cases drawn from de-identified notes in credentialed-access sources such as MIMIC-IV [16] and the n2c2 data [15], plus synthetic notes written by pharmacists to cover rare situations. Two pharmacists will label each case independently for interactions, contraindications and renal-dosing problems, and a third will settle disagreements; agreement will be reported. The knowledge base will combine DrugBank interaction data [17], drug and condition codes from RxNorm [18] and SNOMED CT, and renal-dosing rules from published guidelines, written as logic rules (for example in SWI-Prolog [19]). Every entry records its source. The knowledge base also records which drugs and conditions have been reviewed, so that "no known risk" can be told apart from "knowledge base silent."

2. **Procedure:**

   2.1. **Baselines:** The same LLM will be run without verification in two ways: reading the note and answering directly, and answering with retrieved knowledge-base entries (RAG). The best of these neural systems sets the accuracy reference.

   2.2. **Proposed pipeline:** The LLM extracts structured facts (drugs, doses, conditions, lab values such as eGFR), each linked to the exact evidence span in the note and mapped to standard codes. It then proposes safety conclusions. A symbolic engine tries to derive each conclusion from the extracted facts and the knowledge base, and the derivation is saved as a proof, called a certificate.

   2.3. **Verification and gating:** An independent proof checker validates every step of each certificate, and a span checker confirms that each premise matches the text it cites. Each output then falls into one of three states: risk found (with a certificate), no known risk (the relevant part of the knowledge base was checked and nothing was found), or abstain (the knowledge base is silent or no valid proof exists). Claims without a valid certificate are not shown as safety claims.

2.4. **Pharmacist study:** Licensed pharmacists will review recommendations either with certificates or with free-text LLM rationales, in counterbalanced order. Some recommendations will contain deliberately injected errors, and pharmacists will not be told which ones.

3. **Measurement tools:** Data will be collected with the following measures:

3.1. **Safety and accuracy:** unsupported-claim rate (claims that cannot be derived from the note and the knowledge base), coverage, abstention rate, and precision, recall and F1 for interaction and contraindication detection.

3.2. **Grounding and faithfulness:** grounding rate (share of premises in accepted certificates that point to a correct evidence span) and a perturbation-based faithfulness test. In this test we change or remove a fact that an explanation cites and check whether the conclusion changes accordingly; for LLM rationales we adapt comprehensiveness and sufficiency measures [20].

3.3. **Human-study measures:** error-detection rate, rate of accepting incorrect recommendations, time per case and self-rated confidence, together with short semi-structured interviews about pharmacists' experience.

4. **Data Analysis:** We plan to analyse the two streams of data both independently and together:

Quantitative Analysis: For H1 and H2 we will compare gated and ungated systems on the same cases using paired bootstrap confidence intervals, and test H2 for non-inferiority against the 2-point F1 margin. For H3 we will report the grounding rate with an exact binomial confidence interval against the 95% target and compare faithfulness scores using paired Wilcoxon signed-rank tests. For H4 we will use mixed-effects logistic regression with random effects for pharmacist and case, comparing explanation types on error detection and on acceptance of incorrect recommendations.

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Qualitative Analysis: Interview transcripts and failure cases will be analysed using Braun and Clarke's method of thematic analysis [21]. Each failure will be coded by where it arose: extraction error, code-mapping error, knowledge-base gap, rule-encoding error, or proof/span-check mismatch. This will show where the neural-symbolic interface breaks.

Triangulation: Finally, we will integrate the findings from both datasets, for example by checking whether the failure types found in the qualitative analysis explain the losses in accuracy and coverage, and whether pharmacists' comments agree with the error-detection results. This will strengthen the credibility and validity of our conclusions.

5. **Ethical Considerations:** Only de-identified or synthetic notes will be used, under the data-use agreements and training required by each source, and the knowledge-base licences (for example DrugBank) will be followed. The system is a research prototype and will not be used for real patient care. Pharmacists will consent to the main ethical principles in research with

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human participants, including: informed consent, protection from harm, right to withdraw, confidentiality, debriefing and limited deception (they will not be told in advance which recommendations contain injected errors, and this will be explained in the debriefing). Ethics approval will be sought from the relevant institutional review body before the pharmacist study begins.

Expected Outcome: We anticipate that requiring certificates will remove a large share of unsupported safety claims at only a small loss in F1 and a moderate abstention rate, that most premises in accepted certificates will trace back to real text in the note, and that pharmacists will catch more injected errors with certificates than with free-text rationales. We also expect most failures to occur at the interface (misread notes, wrong drug codes and knowledge-base gaps) rather than in the proof step itself.

Potential Limitation: A certificate only guarantees that a conclusion follows from the extracted facts and the knowledge base. It does not guarantee that the facts were read correctly or that the

knowledge base is complete and correct. Because our knowledge base will cover a limited set of drugs, conditions and rules, the results will offer deep insight into this scope but cannot be applied to all of medication safety or to other clinical tasks. The benchmark uses English notes from a few sources, and the pharmacist study has a modest sample, so its findings are exploratory.

Conclusion: This project is designed as a rigorous, falsifiable test, not as a claim that hallucination is solved, to determine whether letting an AI system speak on medication safety only when it can prove its claim is a workable trade-off between safety, accuracy and usefulness for pharmacists.

Project Practicalities: Our team will work with two to three practising pharmacists to encode rules, annotate the benchmark and take part in the review study, using open-source tools (an LLM, SWI-Prolog and a small independent proof checker). The work is planned in four phases over about 12 months: knowledge base and framework (months 1–3), benchmark annotation (months 3–6), automated evaluation (months 6–9), and the pharmacist study with analysis (months 9–12).

Post-program Plan: Following the analysis, we plan to release the framework, the rule set and the benchmark (where data-use terms allow) as open-source resources with a straightforward guide, and to share our findings, including a written failure analysis, with participating pharmacists, hospital pharmacy departments and the clinical-AI community, hoping they find it useful enough to try and adapt.

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